Cranial Nerves (V, VII, IX, X, XII): Set 5

1. A patient presents with a highly aggressive glomus jugulare tumor that has completely destroyed the structures passing through the jugular foramen. Which clinical presentation will this patient display?
a. Complete ptosis, a fully dilated pupil, and an inability to abduct the eye
b. Trismus, loss of sensation over the lower jaw teeth, and mandible deviation
c. Loss of taste on the posterior tongue, sagging soft palate with a deviated uvula, and paralysis of the trapezius muscle
d. Absolute loss of taste and general somatic sensation over the anterior tongue
e. Total facial paralysis accompanied by severe dryness of the eyes and mouth
2. A 38-year-old patient undergoes an extensive surgical resection of a tumor in the submandibular triangle. Postoperatively, when asked to protrude their tongue, the tongue tip deviates sharply to the right side, and fasciculations are noted on that side. This isolates iatrogenic injury to the:
a. Left glossopharyngeal nerve
b. Left lingual nerve
c. Right hypoglossal nerve (CN XII)
d. Right glossopharyngeal nerve
e. Left vagus nerve trunk
3. A neurosurgeon evaluates a patient with an intracranial lesion compressing the motor root of the trigeminal nerve (CN V). When the patient is asked to open their mouth wide against resistance, the mandible deviates characteristically to the injured side due to the unopposed action of the healthy contralateral:
a. Masseter muscle
b. Temporalis muscle posterior horizontal fibers
c. Medial pterygoid muscle sling
d. Lateral pterygoid muscle
e. Anterior belly of the digastric
4. A patient presents with a severe skull base fracture passing directly through the internal acoustic meatus. Alongside unilateral facial paralysis, the patient experiences severe hyperacusis. The hyperacusis confirms that the lesion is located proximal to the branching of the facial nerve to which muscle?
a. Tensor tympani muscle
b. Tensor veli palatini
c. Stapedius muscle
d. Masseter muscle
e. Lateral rectus muscle
5. A patient experiences an extensive acoustic neuroma at the cerebellopontine angle that completely compresses the facial nerve at its intracranial takeoff. Which secretory deficit will this patient manifest?
a. Loss of parotid gland secretion exclusively
b. Profuse, uncontrollable salivation from all major glands
c. Loss of lacrimation (dry eye) via the greater petrosal nerve, along with reduced submandibular and sublingual salivation via the chorda tympani
d. Loss of sweat secretion over the skin of the forehead only
e. Total hypersecretion of the minor palatine glands

6. Let's briefly touch on another nerve. A patient presents with a selective lesion of the trochlear nerve (CN IV). The patient complains of diplopia (double vision) that makes it exceptionally difficult to perform which routine physical task?
a. Look straight up toward the ceiling
b. Abduct the eye laterally past the midline
c. Close the eye tightly against a strong wind
d. Walk down stairs or read a book (gaze directed downward and inward)
e. Dilate the pupil in a low-light environment
7. A 52-year-old patient presents with agonizing, sharp paroxysms of electric shock-like pain triggered by eating or washing the face, with the pain mapping cleanly over the maxilla skin. This presentation is classic for trigeminal neuralgia affecting which division?
a. Ophthalmic division (V1)
b. Mandibular division (V3)
c. Maxillary division (V2)
d. Chorda tympani nerve
e. Greater petrosal nerve
8. During an extraoral surgical repair of a mandibular fracture near the angle of the jaw, a clinician accidentally cuts the marginal mandibular branch of the facial nerve. This iatrogenic injury will manifest as a complete inability to:
a. Wrinkle the forehead skin
b. Close the eye tightly against resistance
c. Compress the cheek against the lower molars
d. Depress the corner of the mouth on the affected side, causing an asymmetrical smile
e. Protrude the lower jaw forward
9. A patient presents with a deep tumor expanding within the infratemporal fossa. Clinical examination reveals a total loss of general somatic sensation (pain, touch) over the anterior two-thirds of the tongue and floor of the mouth on that side, but their taste perception in that exact area is completely normal. This isolates a lesion of the:
a. Chorda tympani nerve proximal to joining the lingual nerve
b. Lingual nerve *proximal* to its junction with the chorda tympani
c. Lingual nerve *distal* to its junction with the chorda tympani
d. Glossopharyngeal nerve trunk
e. Hypoglossal nerve main stem
10. A 44-year-old patient undergoes a deep parotidectomy. Postoperatively, when the patient smells or eats food, the skin over the parotid region beads with sweat and turns red (Frey's syndrome). This occurs because regenerated parasympathetic fibers from the otic ganglion mistakenly grew to innervate local sweat glands. These fibers originally traveled via the:
a. Chorda tympani nerve
b. Greater petrosal nerve
c. Long buccal nerve
d. Auriculotemporal nerve
e. Greater occipital nerve
11. A neurosurgeon performs a precise evaluation of a patient with an upper motor neuron (UMN) lesion of the facial nerve following a cortical stroke in the left hemisphere. This patient will present with:
a. Total paralysis of the entire left side of the face

b. Total paralysis of the entire right side of the face
c. Paralysis of only the lower quadrant of the face on the right side, with normal forehead wrinkling and eye closure on both sides
d. Sparing of the lips, with complete loss of eye closure bilaterally
e. Complete loss of taste over the entire tongue, with normal muscle function everywhere

12. Forehead wrinkling and eye closure are completely spared in an upper motor neuron facial nerve lesion because the motor neurons in the facial nucleus that supply the upper face receive:

a. Bilateral corticobulbar innervation from both cerebral hemispheres
b. Direct supply from the cervical spinal cord segments
c. Only ipsilateral cortical tracks
d. No cortical representation at all
e. Motor control via the trigeminal nerve mesencephalic tract

13. A patient presents with a rare condition called glossopharyngeal neuralgia, featuring severe paroxysms of pain triggered by swallowing. The pain maps to the tonsillar fossa and deep ear. To interrupt this pain pathway, a surgeon can selectively target the glossopharyngeal nerve, whose somatic sensory fibers carry pain signals to terminate inside which brainstem nucleus?

a. Nucleus ambiguus
b. Nucleus solitarius gustatory quadrant
c. Spinal nucleus of the trigeminal nerve (CN V)
d. Dorsal motor nucleus
e. Edinger-Westphal nucleus

14. A physician initiates a gag reflex test by touching the posterior oropharyngeal mucosa with a tongue depressor. The patient displays a normal, rapid contraction of the pharyngeal muscles. The efferent motor limb of this protective reflex is mediated by SVE fibers traveling within the:

a. Trigeminal nerve (V3)
b. Facial nerve (CN VII)
c. Glossopharyngeal nerve (CN IX)
d. Vagus nerve (CN X via the pharyngeal plexus)
e. Hypoglossal nerve (CN XII)

15. A patient presents with a severe neuro-neoplastic infiltration of the pterygopalatine fossa. This lesion completely blocks the pterygopalatine ganglion, resulting in a total loss of secretomotor function to the:

a. Parotid salivary gland
b. Lacrimal gland and the minor mucous glands of the nose and palate
c. Submandibular and sublingual glands
d. Sublingual gland exclusively
e. Glands of Von Ebner

16. A patient presents with a complete unilateral lower motor neuron lesion of the vagus nerve (CN X). When the patient is asked to open their mouth and say "Ah", physical evaluation reveals a sagging soft palate on the injured side and a deviated uvula that points toward:

a. The paralyzed, injured side
b. The floor of the mouth vertically
c. The normal, healthy side
d. The incisive foramen anteriorly

e. The epiglottis fold

17. Let's do a brief touch on another cranial nerve. A patient presents with a complete inability to abduct the right eye laterally past the midline, resulting in a permanent internal strabismus. This isolates a lower motor neuron lesion of which cranial nerve?

a. Right oculomotor nerve (CN III)
b. Right trochlear nerve (CN IV)
c. Right ophthalmic nerve (V1)
d. Right abducens nerve (CN VI)
e. Right facial nerve (CN VII)

18. A patient experiences an expanding aneurysm of the internal carotid artery within the cavernous sinus. This vascular lesion compresses the abducens nerve (CN VI) as it tracks through the center of the sinus. The patient will present with an inability to:

a. Close the eye tightly
b. Open the mouth against resistance
c. Sense touch on the forehead skin
d. Abduct the eyeball laterally
e. Protrude the tongue straight

19. A 62-year-old patient undergoes a radical thyroidectomy within the muscular triangle. Postoperatively, the patient experiences a distinct change in vocal quality characterized by severe, persistent hoarseness due to iatrogenic injury to the:

a. Superior laryngeal nerve internal branch
b. External laryngeal nerve supplying cricothyroid
c. Recurrent laryngeal nerve (branch of CN X)
d. Pharyngeal branch of vagus
e. Hypoglossal nerve trunk

20. If a surgeon accidentally damages the external laryngeal nerve while ligating the superior thyroid artery at the upper pole of the thyroid gland, the patient will lose the function of which single muscle, resulting in an inability to create high-pitched vocal sounds?

a. Vocalis muscle
b. Lateral cricoarytenoid
c. Thyroarytenoid
d. Cricothyroid muscle
e. Posterior cricoarytenoid

21. A patient presents with an emergency condition where a sharp fishbone is stuck deep within the piriform recess of the laryngopharynx. During attempts to retrieve the bone extraorally, a clinician must be careful to avoid injuring a nerve that runs immediately beneath the thin mucosa of the recess, which is the:

a. External laryngeal nerve
b. Recurrent laryngeal nerve
c. Glossopharyngeal nerve trunk
d. Internal laryngeal nerve (branch of CN X)
e. Hypoglossal nerve root

22. Direct traumatic injury to the internal laryngeal nerve within the piriform recess will result in a complete loss of which vital protective reflex mechanism?

a. Pupillary light reflex
b. Corneal blink reflex

c. Cough reflex (due to loss of sensory awareness of the larynx mucosa *superior* to the vocal folds)
d. Jaw-jerk reflex
e. Salivary secretion reflex

23. During a surgical procedure inside the carotid triangle, a surgeon accidentally transects the superior root of the ansa cervicalis (descendens hypoglossi). This will result in a loss of motor tone and subsequent paralysis of which group of neck muscles?

a. Suprahyoid muscles
b. Muscles of mastication
c. Extrinsic tongue muscles
d. Infrahyoid strap muscles (except thyrohyoid)
e. Intrinsic laryngeal muscles

24. All infraheard strap muscles are innervated by the ansa cervicalis loop, with one selective exception that is supplied directly by C1 fibers hitchhiking along the hypoglossal nerve. This exception is the:

a. Sternohyoid muscle
b. Sternothyroid muscle
c. Thyrohyoid muscle
d. Omohyoid muscle superior belly
e. Cricothyroid muscle

25. A patient experiences a severe fracture through the stable body of the mandible passing through the mandibular canal. This trauma will selectively compress or tear the inferior alveolar nerve, leading to complete anesthesia over the:

a. Anterior two-thirds of the tongue
b. Mucosa of the hard palate
c. Skin of the lower lip and chin (via the mental nerve)
d. Mucosa of the soft palate and uvula
e. Vestibular gingiva of maxillary molars

26. When evaluating the corneal reflex, a cotton wisp is touched to the left cornea. The patient displays a normal blink response in the right eye (consensual), but the left eye remains completely still. This specific pattern isolates a lower motor neuron lesion of the:

a. Left ophthalmic nerve (V1)
b. Right ophthalmic nerve (V1)
c. Left facial nerve (CN VII motor branches to orbicularis oculi)
d. Right facial nerve (CN VII)
e. Left oculomotor nerve (CN III)

27. A patient experiences an isolated lesion of the long buccal nerve (a branch of V3) due to a deep laceration in the cheek. Clinical evaluation will reveal a loss of:

a. Motor control of the buccinator muscle
b. General somatic sensation (pain, touch) from the skin and mucous membrane of the cheek
c. Secretomotor drive to the parotid gland
d. Taste sensation from the anterior tongue
e. Motor control of the masseter muscle

28. The tensor tympani muscle works to pull the handle of the malleus medially, tensing the tympanic membrane to protect the ear from loud sounds. It develops from the first pharyngeal arch and is innervated by which nerve?
a. Facial nerve (CN VII)
b. Glossopharyngeal nerve (CN IX)
c. Mandibular nerve (V3 division of Trigeminal)
d. Chorda tympani nerve
e. Vagus nerve (CN X)
29. The tensor veli palatini muscle works to tense the soft palate during swallowing. This muscle develops from the first pharyngeal arch and receives its motor innervation from a branch of the:
a. Facial nerve (CN VII)
b. Maxillary nerve (V2)
c. Mandibular nerve (V3 division of the Trigeminal nerve)
d. Glossopharyngeal nerve (CN IX)
e. Vagus nerve (CN X)
30. A patient presents with an inability to tighten or elevate the soft palate on the left side, causing a palatal drop, but their jaw movements and facial expressions are completely normal. This isolates a lower motor neuron lesion of the pharyngeal branches of the:
a. Left trigeminal nerve (V3)
b. Left facial nerve (CN VII)
c. Left vagus nerve (CN X via the pharyngeal plexus)
d. Left hypoglossal nerve (CN XII)
e. Left spinal accessory nerve
31. Digital pressure applied over the carotid sinus baroreceptors within the carotid triangle can cause a profound physiological drop in heart rate and blood pressure. The afferent pathway of this baroreceptor reflex is carried by which nerve branch?
a. Carotid sinus nerve of Hering (branch of CN IX)
b. Superior laryngeal nerve (branch of CN X)
c. Recurrent laryngeal nerve
d. Transverse cervical nerve
e. Sympathetic trunk rami
32. The jugulodigastric lymph node is a large, clinically prominent deep cervical node located at the intersection of the posterior belly of the digastric muscle and the internal jugular vein. Enlargement and tenderness of this specific node primarily indicates an infection or malignancy originating within the:
a. Apex of the chin
b. Palatine tonsil or posterior tongue
c. Lower incisor teeth
d. Skin over the occipital bone
e. Sublingual salivary gland
33. If the spinal accessory nerve (CN XI) is completely severed inside the posterior triangle of the neck during a lymph node biopsy, the patient will experience a complete inability to:
a. Rotate the head to the opposite side
b. Shrug the shoulder or abduct the arm past the horizontal plane due to trapezius paralysis
c. Protrude the tongue straight

d. Depress the mandible to open the mouth
e. Swallow solid food chunks

34. Let's briefly touch on another cranial nerve. A patient presents with an inability to constrict the pupil (mydriasis) and a complete loss of the accommodation reflex for near vision. This identifies a loss of parasympathetic tracking inside which cranial nerve?

a. CN II (Optic)
b. CN III (Oculomotor nerve)
c. CN IV (Trochlear)
d. CN VI (Abducens)
e. CN V1 (Ophthalmic)

35. A 7-year-old child presents with severe acute otitis media that has caused inflammation of the chorda tympani nerve as it runs across the tympanic membrane. The child will manifest a loss of:

a. General somatic pain sensation on the tongue tip
b. Taste sensation over the anterior two-thirds of the tongue on that side
c. Complete paralysis of the masseter muscle
d. Ptosis of the upper eyelid
e. Deafness in both ears

36. A patient undergoes an extensive radical neck dissection for oral cancer. Postoperatively, the patient experiences a complete loss of general somatic sensation affecting the anterior two-thirds of the tongue on that side due to iatrogenic damage to the:

a. Posterior superior alveolar nerve
b. Lingual nerve (branch of V3)
c. Glossopharyngeal nerve (CN IX)
d. Mylohyoid nerve
e. Vagus nerve trunk

37. The general somatic sensation (pain and touch) from the skin of the external acoustic meatus and tympanic membrane is shared by branches of the trigeminal nerve and the auricular branch of the:

a. Facial nerve
b. Vagus nerve (Arnold's nerve)
c. Glossopharyngeal nerve
d. Hypoglossal nerve
e. Accessory nerve

38. Stimulation of the auricular branch of the vagus nerve (Arnold's nerve) by a speculum inside the external ear canal can occasionally trigger reflex coughing or fainting due to:

a. Primary activation of the tongue muscles
b. Reflex vagal bradycardia and parasympathetic activation
c. Sudden high-pressure arterial flows
d. Intermittent spasms of the masseter muscle
e. Direct stimulation of the olfactory bulb

39. A patient presents with a deep skull base tumor that completely compresses the mandibular nerve (V3) immediately as it exits the foramen ovale. Which clinical presentation will be observed regarding the secretomotor function of the parotid salivary gland?

a. Complete loss of parotid secretion due to destruction of preganglionic fibers
b. Profuse, uncontrollable salivation from Stensen's duct
c. Normal parotid secretion because its parasympathetic fibers travel via the facial nerve
d. Loss of parotid secretomotor drive because the postganglionic fibers travel via the auriculotemporal nerve (a branch of V3)
e. Total hypertrophy of the gland parenchyma

40. A patient presents with severe, sharp paroxysms of stabbing pain triggered by eating or shaving, with the pain radiating along the mandibular body skin. This diagnostic condition is:

a. Bell's palsy
b. Trigeminal neuralgia (Tic douloureux affecting V3)
c. Glossopharyngeal neuralgia
d. Frey's syndrome
e. Horner's syndrome